

for Nanocrystalline Composites

Basic Model Configuration

Advanced Model Configuration

Temperature:

Temperature Gradient:

Silicon Doping:

Germanium Doping:

Substrate Composition Si(x)Ge(1-x):

Mat 1 Material:

Mat 2 Material:

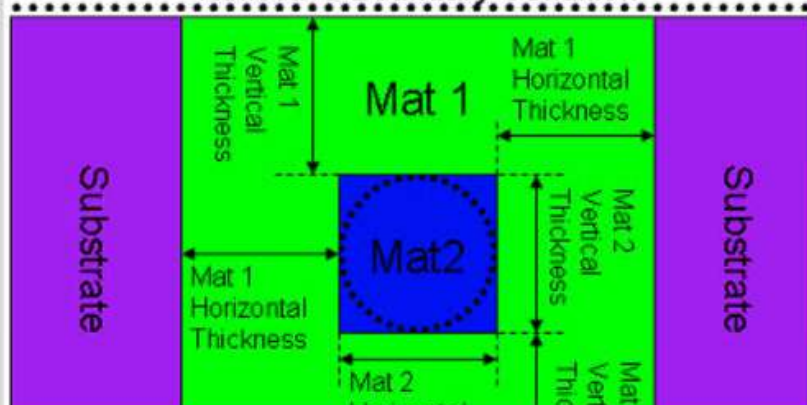
Mat 1 Horizontal Thickness:

Mat 1 Vertical Thickness:

Mat 2 Horizontal Thickness:

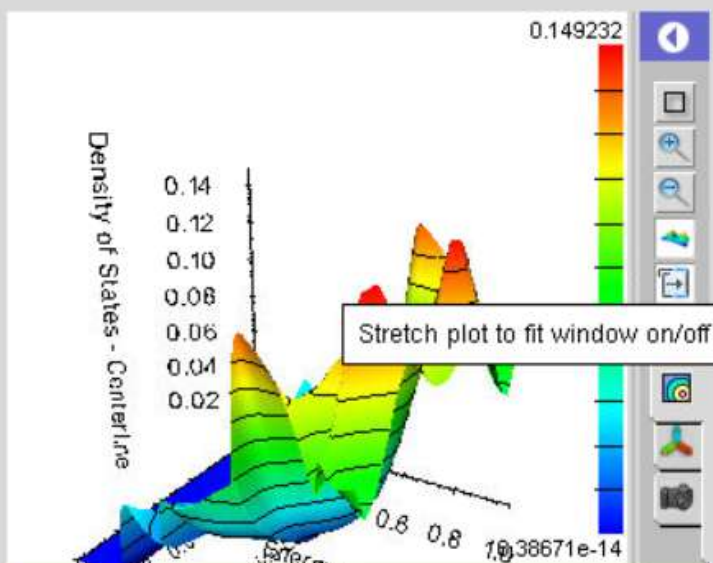
Mat 2 Vertical Thickness:

Reflective Boundary Condition



Simulate

Result:



2 results

Clear One

Clear All

All

Simulation = #2

Mat 1 Horizontal Thickness = 0.2nm

Mat 1 Vertical Thickness = 0.2nm

Mat 2 Horizontal Thickness = 0.2nm

Mat 2 Vertical Thickness = 0.2nm

barrierLx = 0.2nm

barrierLy = 0.2nm

wellLx = 0.2nm

wellLy = 0.2nm

Storage (manage)

0% of 10.000004GB



780 × 600

Thermoelectric Power Factor Calculator
for Nanocrystalline Composites

Terminate Keep for later

Basic Model Configuration Advanced Model Configuration

Temperature: 300K

Temperature Gradient: 10K

Silicon Doping: 1e+18/cm3

Germanium Doping: 1e+18/cm3

Substrate Composition Si(x)Ge(1-x): 1

Mat 1 Material: Silicon

Mat 2 Material: Germanium

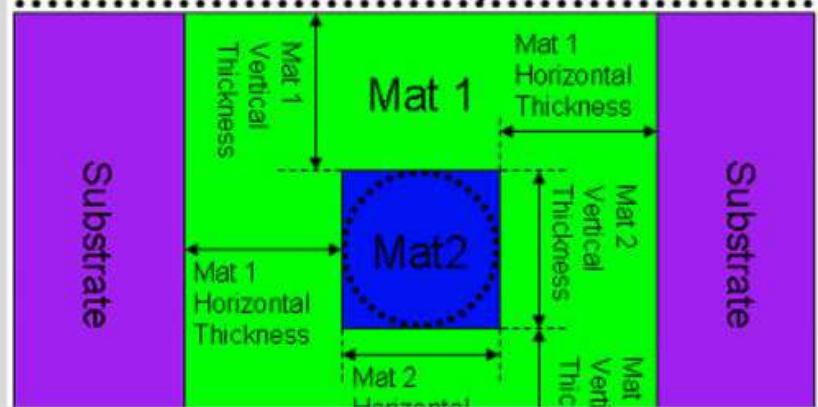
Mat 1 Horizontal Thickness: 0.5nm

Mat 1 Vertical Thickness: 0.5nm

Mat 2 Horizontal Thickness: 0.5nm

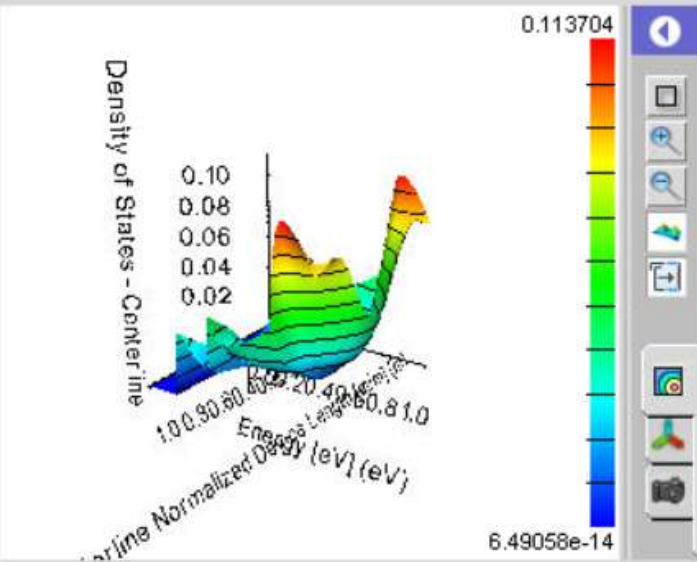
Mat 2 Vertical Thickness: 0.5nm

Reflective Boundary Condition



Simulate

Result: Density of States - Centerline



2 results

Clear One

Clear All

All

Simulation = #2

Mat 1 Horizontal Thickness = 0.5nm
Mat 1 Vertical Thickness = 0.5nm
Mat 2 Horizontal Thickness = 0.5nm
Mat 2 Vertical Thickness = 0.5nm
barrierLx = 0.5nm
barrierLy = 0.5nm
wellLx = 0.5nm
wellLy = 0.5nm

for Nanocrystalline Composites

Basic Model Configuration

Advanced Model Configuration

Temperature:

Temperature Gradient:

Silicon Doping:

Germanium Doping:

Substrate Composition Si(x)Ge(1-x):

Mat 1 Material:

Mat 2 Material:

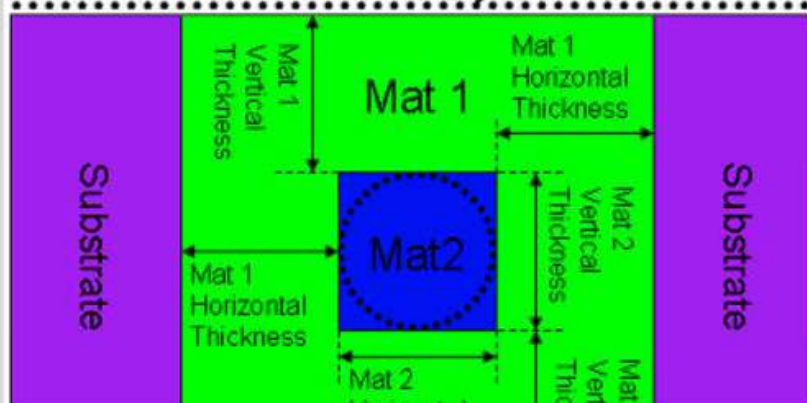
Mat 1 Horizontal Thickness:

Mat 1 Vertical Thickness:

Mat 2 Horizontal Thickness:

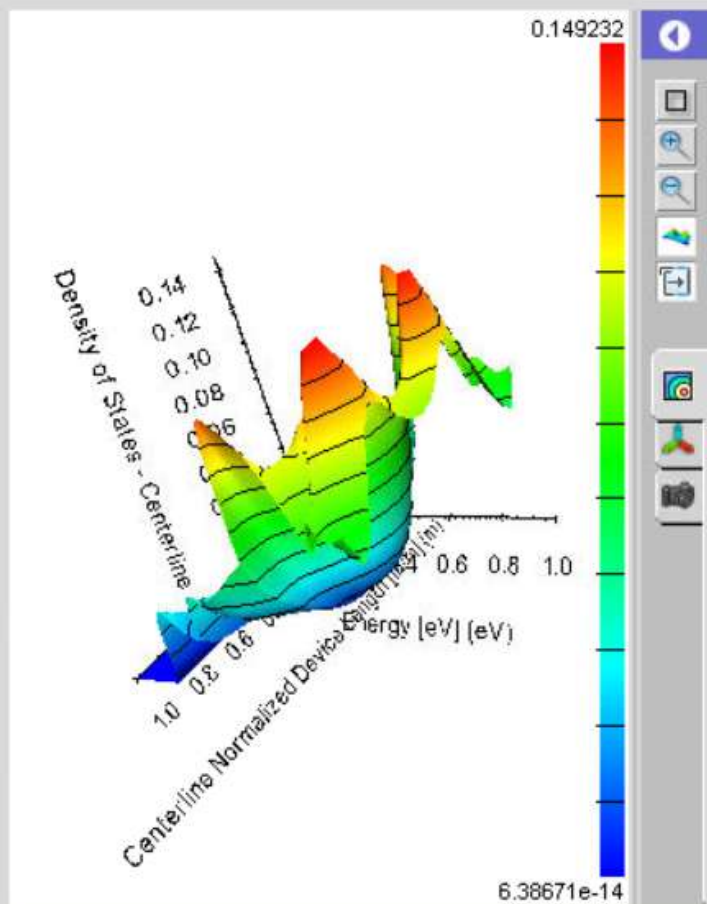
Mat 2 Vertical Thickness:

Reflective Boundary Condition



Simulate

Result:



1 result

Clear

Storage (manage)

0% of 10.000004GB



780 × 600